Tools and APIs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Teach both sides of the tool boundary.

Supply full schemas, distinguish missing from zero and keep operation permission separate from valid data.

Keep one habit: Ask whether the request was permitted, whether it ran, and what the returned evidence actually supports.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Explain the **mechanism** and terminology of tools and **api contract**s

Two 45-minute sessions. Tools expose structured operations with **input** and **output** contracts. Generating a **tool call** is a proposal; validation and authorisation decide whether it executes.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

By the end, I can explain tools and apis using a traced example and a stated limitation.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Lesson and workbook; paper and pencil
- Calculator and optional local Python 3 for arithmetic; no live API or private data
- Prepare the fresh retry separately and withhold its solution until after collection

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Conversation State and Memory. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of conversation state and retention

Reference example, not the only acceptable wording: Teacher explanation: Conversation state stores facts or prior interactions for later use.

It needs scope, provenance, correction and retention rules rather than indefinite accumulation. This classroom assistant keeps an approved room ID only for the current session.

A later correction replaces the active value but retains an audit event until session end. Do not persist private text merely because it appeared in conversation.

Without opening the old lesson, explain one key idea from Retrieval Augmented Generation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of building a grounded answer

Reference example, not the only acceptable wording: Teacher explanation: A grounded answer connects each factual claim to **evidence** actually retrieved for the task.

Retrieval success does not automatically make generation faithful. Build a claim-**evidence** table before final wording.

Preserve source versions and qualifiers, omit unsupported claims or explicitly state the gap, then validate the answer against the table.

Without opening the old lesson, explain one key idea from Source Quality. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of conflicting and time-sensitive sources

Reference example, not the only acceptable wording: Teacher explanation: Changing facts require a stated time and source-version policy. Current-looking records can conflict without enough **evidence** to choose between them.

For this toy task a signed amendment supersedes the referenced original from its effective time. Keep both records. A newer timestamp alone has no authority unless the policy grants it.

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

▼ **Earlier wording, alternatives and picture transcripts**

Teach the learning cycle

Retrieval answer guidance

Conversation State and Memory

Retrieval Augmented Generation

Source Quality

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A tool can calculate or search supplied notes. The request is 'What is 17×6, and where do roots take up water?' Assign each subtask and describe **output** checks.

Concept to explain

A **tool call** is a structured request to an external capability with returned data.

Term	Meaning
API	An agreed interface through which software requests data or actions from another system.
tool call	A structured request to a tool with specified arguments.
argument	A value supplied to a function or tool to tell it what to operate on.
error response	A message or structured result indicating that a request could not be completed as expected.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach proposal → validation → operation → checked response

> Keep the room and book contracts separately labelled.

State the room contract

Toy lookup_room contract	Declared rule
Operation and permission	Read-only lookup of this public classroom registry; no booking or deletion
Request shape	Exactly one field room_id; nonempty string; no unknown fields
Registry	B has capacity 12; Z has measured capacity 0; all other room IDs absent
Successful result	found=true with matching room_id and exact integer capacity>=0
Missing result	found=false with matching room_id; no capacity measurement
Invalid request	Reject before invocation; do not silently remove extra operations
Transient timeout	At most2 total attempts including first, for this read-only operation only; then report unavailable
Do not retry	Invalid arguments, not found, denied access or invalid response; no invented success

Tools expose structured operations with **input** and **output** contracts. Generating a **tool call** is a proposal; validation and authorisation decide whether it executes.

For lookup_room accept exactly room_id as a nonempty string; return found plus capacity or a not-found result. It is read-only. Reject unknown **argument**s and do not replace missing values with invented facts.

Trace the room extra-field rejection

Proposed call lookup_room has room_id=B and extra **argument** book=true.

Reject the extra field under the closed **input** contract. The lookup tool provides no booking authority even if the requested room exists.

Read the separate book case

Check a library lookup call before it reaches a tool.

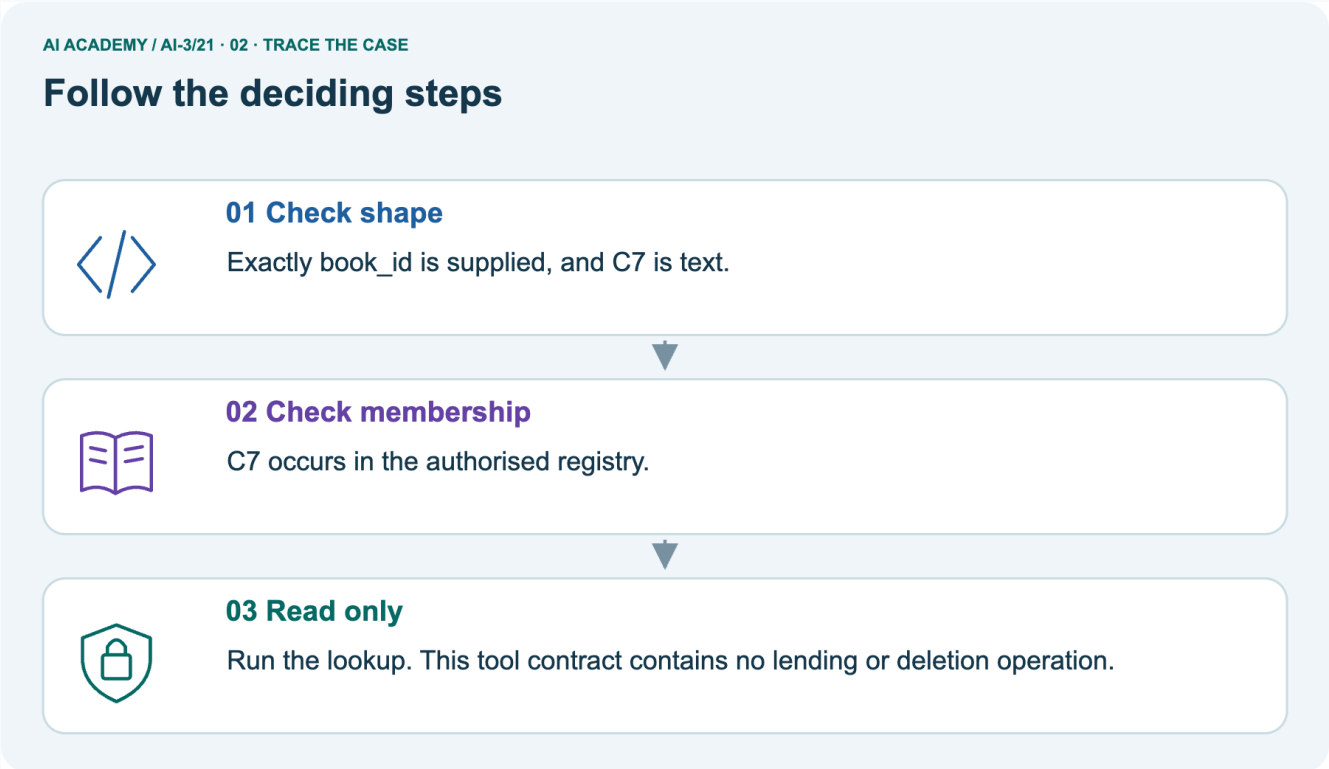
lookup_book accepts exactly one field: book_id, a string. Registry IDs are C7 and M2. The proposed call is {book_id: C7}.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The separate lookup_book example is a membership-and-operation trace: supplied authorised registry IDs are C7 and M2, and the one allowed input field is book_id as text. The pictured call uses C7. No book-content record or actual response is supplied, so this trace establishes an eligible read-only request, not a completed lookup result. The room tool's capacity fields do not belong to the book schema.

Object notations in these diagrams are classroom shorthand; exact JSON syntax is introduced next week.

Trace the operation boundary



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check shape	Exactly book_id is supplied, and C7 is text.
2. Check membership	C7 occurs in the authorised registry.
3. Read only	Run the lookup. This tool contract contains no lending or deletion operation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Run the lookup. This tool contract contains no lending or deletion operation. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict adding lend

Change: Add lend: true to the same call.

Prediction: Reject the call before invoking the tool.

Reason: An extra operation is outside the exact allowed contract, even when book_id is valid.

Name response and access limits

A valid **tool call** can still return unavailable or outdated records. Validate the response and access rights separately.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Additional practice · Choosing calculator and search

An assistant has calculator and source-search tools. Request: 12 times 9 and habitat definition from a class pack.

Choose and verify tools.

Calculator returns 108; search retrieves the definition. Verify **input**s, **output**s and supporting passage before answering.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Tool schema requires operation and two numbers.

2. Request sends operation=divide, a=18, b=24.

3. Tool returns result=0.75; product formats 75%.

4. Timeout or invalid **input** must trigger retry, repair, or safe explanation—not invented data.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Exact explanation to model

Explain the added worked example

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term	Meaning
API contract	Declared input, output and failure rules for an interface
Tool call	A proposed or executed operation with structured arguments
Not found	An explicit absence result rather than a fabricated value

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

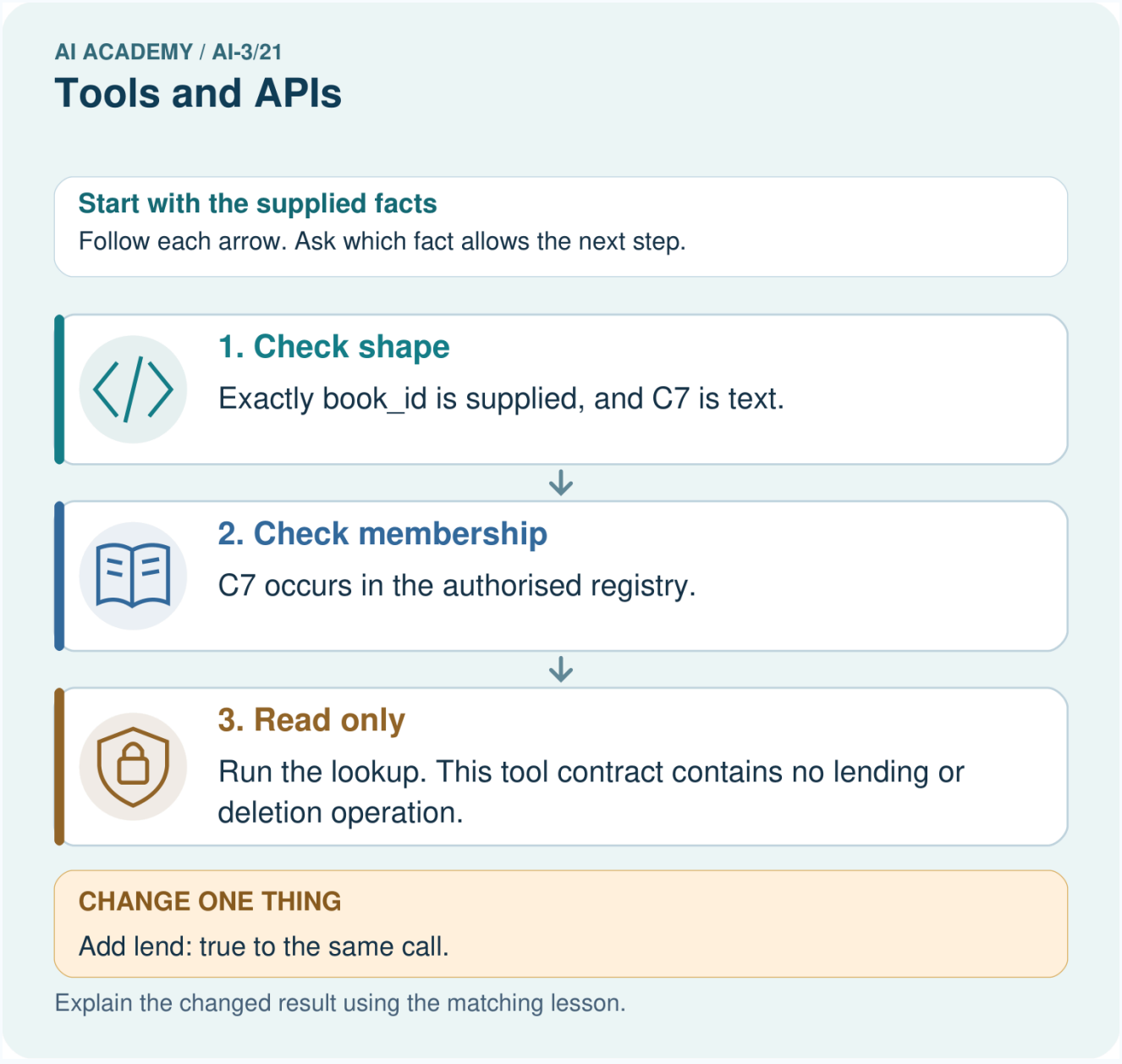
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Tools and APIs

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the explanation, then remove help

> Collect each learner’s reasoning before showing feedback.

Main completion prompt

Completion task: Proposed call lookup_room has room_id=B and extra **argument** book=true. Show the intermediate steps and give the conclusion with its scope.

Expected reasoning and fading help

Teacher check: Teacher explanation: Reject the extra field under the closed **input** contract. The lookup tool provides no booking authority even if the requested room exists.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Tools and **API contract**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A successful tool result supports only its declared data and operation; it does not authorise subsequent side effects.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

Does a valid C7 ID permit the extra lending action? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

No. ID validity and operation permission are separate checks; reject the extra field under this contract. Changed-case check: Reject the call before invoking the tool.

An extra operation is outside the exact allowed contract, even when book_id is valid.

Changed-case reasoning: Reject the call before invoking the tool. An extra operation is outside the exact allowed contract, even when book_id is valid.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: An assistant has calculator and source-search tools. Request: 12 times 9 and habitat definition from a class pack.

Write the question to explain: Choose and verify tools.

Reveal after **prediction**s: Calculator returns 108; search retrieves the definition. Verify **input**s, **output**s and supporting passage before answering.

Use the earlier changed question as optional later practice. Reserve Validate the fresh K room calls independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: Reject the extra field under the closed **input** contract. The lookup tool provides no booking authority even if the requested room exists.

Qualification: A successful tool result supports only its declared data and operation; it does not authorise subsequent side effects.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.

Stage	Teacher action
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

Six-card key: A valid B/12; B valid Z/0; C extra field rejected; D numeric ID rejected; E unavailable after two total timeout attempts; F response identity mismatch rejected. Book trace establishes only eligible request, no supplied actual result. Independent K calls: numeric ID invalid, text K valid input but not found, extra delete field invalid. Not found has no measured capacity; it is not zero.

Run and record

1. Predict → Test → Record → Explain

2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.

3. Use the six supplied toy room-lookup cards and complete closed contract. Validate **argument**s, action scope and response identity; distinguish measured zero from missing and stop after at most two total read-only timeout attempts. Record paper traces, not live calls.

4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

The worked result establishes every future case: A successful tool result supports only its declared data and operation; it does not authorise subsequent side effects.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Validate the fresh K room calls independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: Tool accepts room_id string. Calls are {room_id:7}, {room_id:K} and {room_id:K, delete:true}. Validate; K is absent from the registry.

Teacher explanation: First fails type; second is valid **input** returning **not found**; third fails extra-field validation. **Not found** is not capacity zero.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Tools expose structured operations with **input** and **output** contracts. Generating a **tool call** is a proposal; validation and authorisation decide whether it executes.

For lookup_room accept exactly room_id as a nonempty string; return found plus capacity or a not-found result. It is read-only. Reject unknown **argument**s and do not replace missing values with invented facts.

W2 Trace the worked case

Proposed call lookup_room has room_id=B and extra **argument** book=true. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Reject the extra field under the closed **input** contract. The lookup tool provides no booking authority even if the requested room exists.

W3 Audit the limit

A report omits this qualification: A successful tool result supports only its declared data and operation; it does not authorise subsequent side effects.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A successful tool result supports only its declared data and operation; it does not authorise subsequent side effects.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A successful tool result supports only its declared data and operation; it does not authorise subsequent side effects.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Calculator returns 108; search retrieves the definition. Verify **input**s, **output**s and supporting passage before answering.

Additional transfer practice question

A tool can look up book availability and also place an order. The user asks only whether a book is available. Which action is authorized?

Reasoned practice answer

A read-only availability lookup is authorized. Placing an order is a different action and requires explicit authorization.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Access to a tool automatically authorizes every action it supports.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Calculate 102 and verify 17×6. Retrieve the relevant roots passage and cite its supported answer. Check tool **input**s and returned records. Neither a calculator nor a search tool should be given unrelated action permissions.

Explanations for the additional workbook examples

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Tool accepts room_id string. Calls are {room_id:7}, {room_id:K} and {room_id:K, delete:true}. Validate; K is absent from the registry.

Teacher explanation: First fails type; second is valid **input** returning **not found**; third fails extra-field validation. **Not found** is not capacity zero.

W5 Design a revealing follow-up

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Allow only retrieval and calculator tools with validated **argument**s and visible results.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support while preserving the reasoning goal

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: First fails type; second is valid **input** returning **not found**; third fails extra-field validation. **Not found** is not capacity zero. Require intermediate reasoning and the stated scope.

Fresh retry: Room M exists with measured capacity zero; room N is absent. Distinguish results. Expected reasoning: M found with capacity 0; N **not found** with no measured capacity. Preserve zero separately from missing.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Plan the delayed check

Suggested return points in this level: ai-3/22, ai-3/24, ai-3/27. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 22: Requests, Responses, and JSON. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week 22: Requests Responses and JSON. Carry forward exact input/output contracts and response checks when learning the precise wire-format notation.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Verified Study Buddy

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Explain the **mechanism** and terminology of tools and **api contract**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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Practice and assessment reminders

Before class, solve the supported practice: Proposed call lookup_room has room_id=B and extra **argument** book=true. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Reject the extra field under the closed **input** contract. The lookup tool provides no booking authority even if the requested room exists.

Check every learner before discussion: Use the worked case for Tools and **API contract**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A successful tool result supports only its declared data and operation; it does not authorise subsequent side effects.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Add lend: true to the same call. Require a **prediction** and a reason before comparing. Expected result: Reject the call before invoking the tool.

Reason: An extra operation is outside the exact allowed contract, even when book_id is valid.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

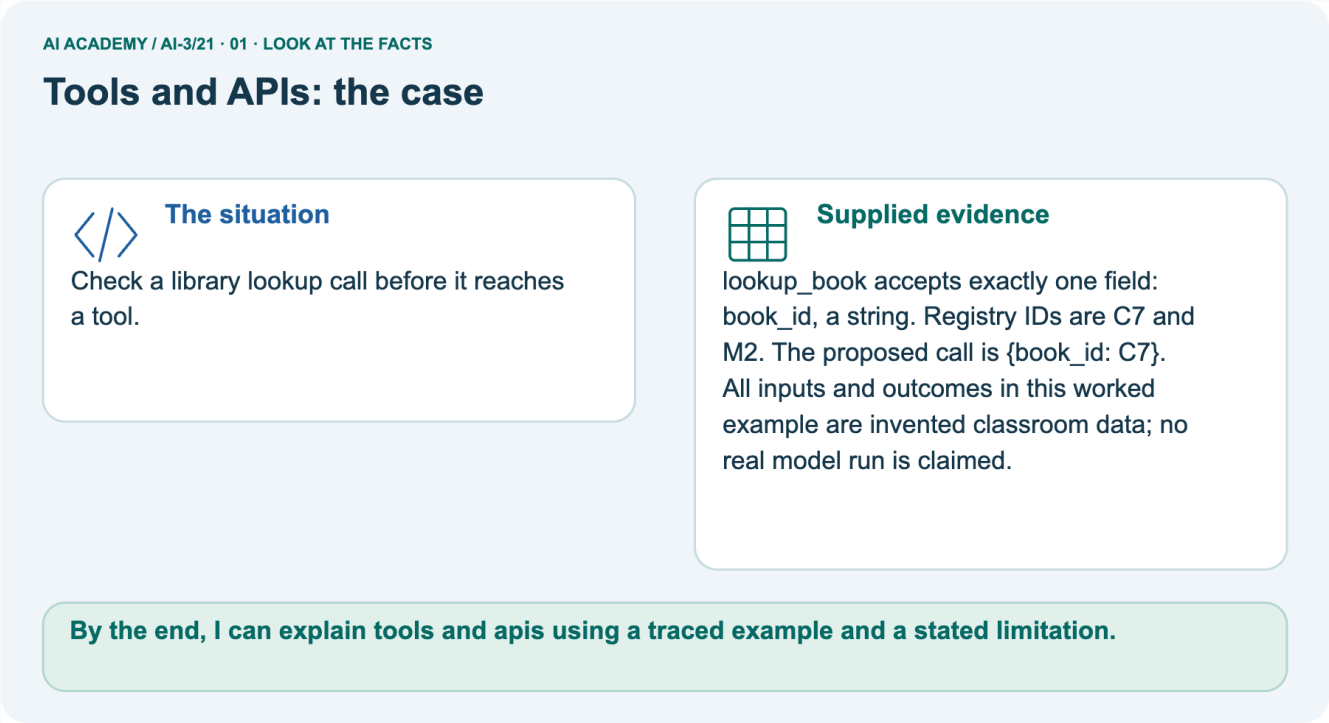
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Tools and APIs: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

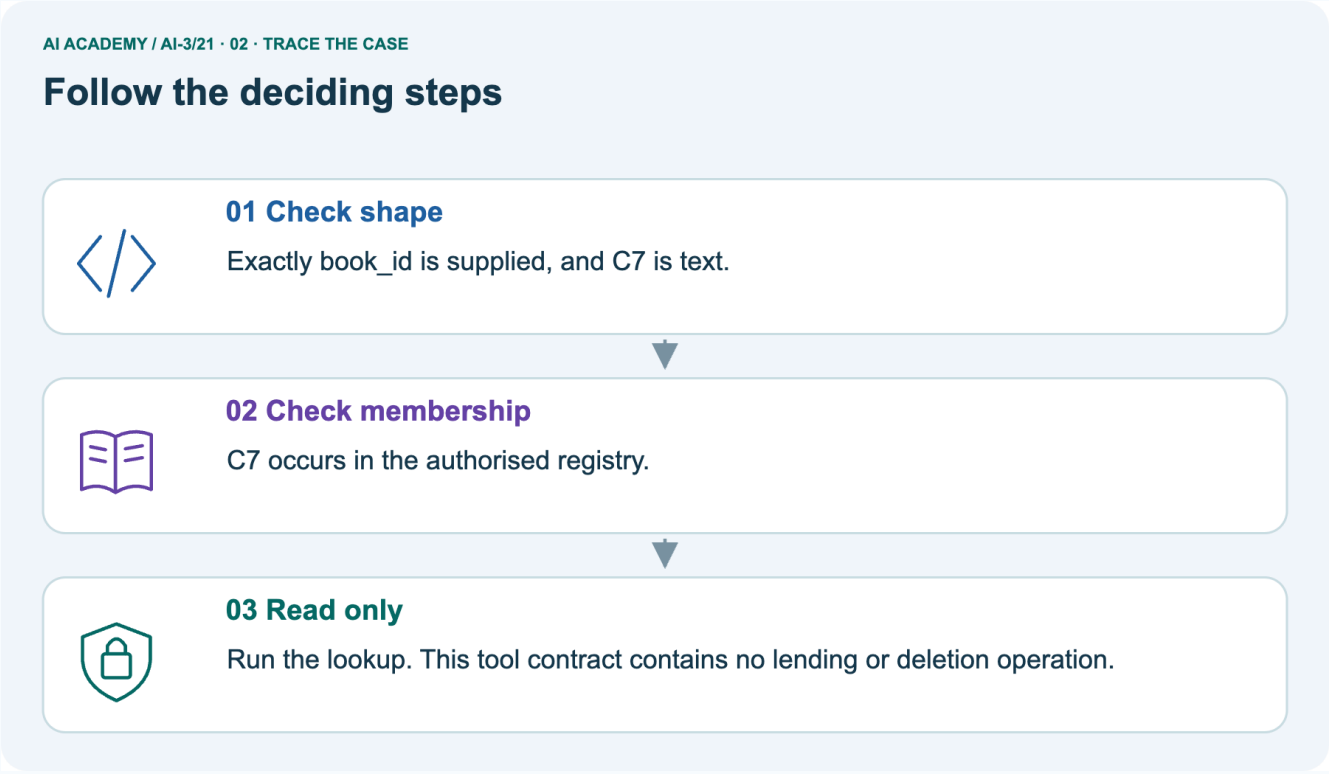
Check a library lookup call before it reaches a tool. Supplied facts: lookup_book accepts exactly one field: book_id, a string. Registry IDs are C7 and M2. The proposed call is {book_id: C7}.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



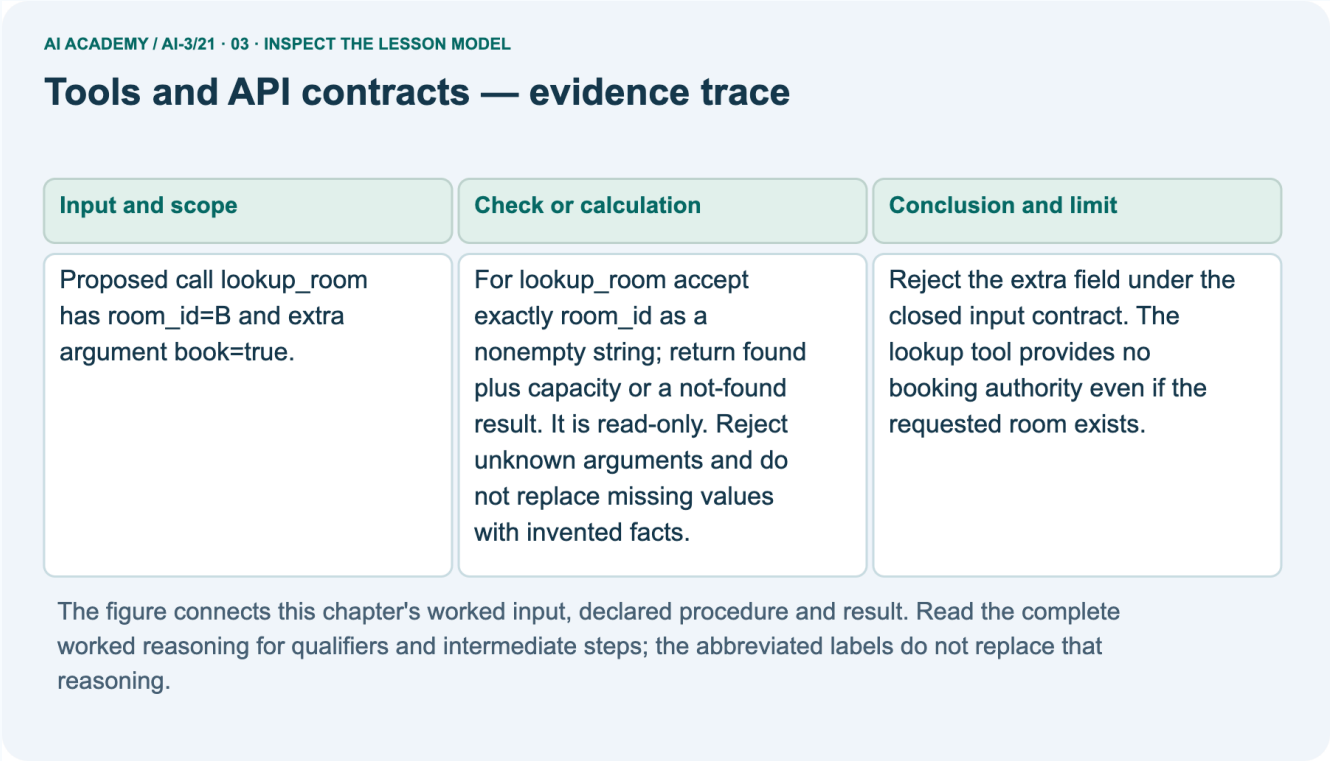
Follow the labelled picture with the matching explanation. Open the vector diagram

Check shape: Exactly book_id is supplied, and C7 is text. → Check membership: C7 occurs in the authorised registry. → Read only: Run the lookup. This tool contract contains no lending or deletion operation.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Tools and API contracts — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Tools and **API contract**s — **evidence** trace. Proposed call lookup_room has room_id=B and extra **argument** book=true. | For lookup_room accept exactly room_id as a nonempty string; return found plus capacity or a not-found result.

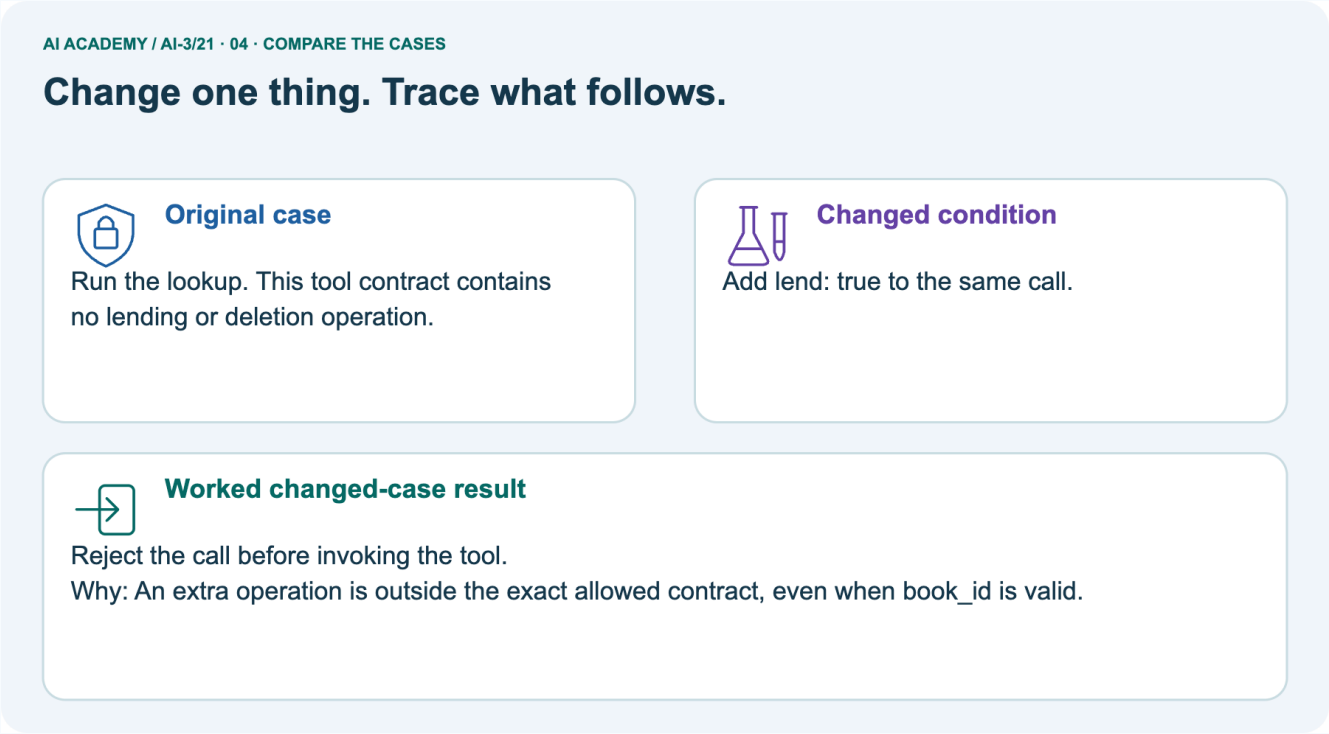
It is read-only. Reject unknown **argument**s and do not replace missing values with invented facts. | Reject the extra field under the closed **input** contract.

The lookup tool provides no booking authority even if the requested room exists.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Run the lookup. This tool contract contains no lending or deletion operation. Changed condition: Add lend: true to the same call. Expected result: Reject the call before invoking the tool.

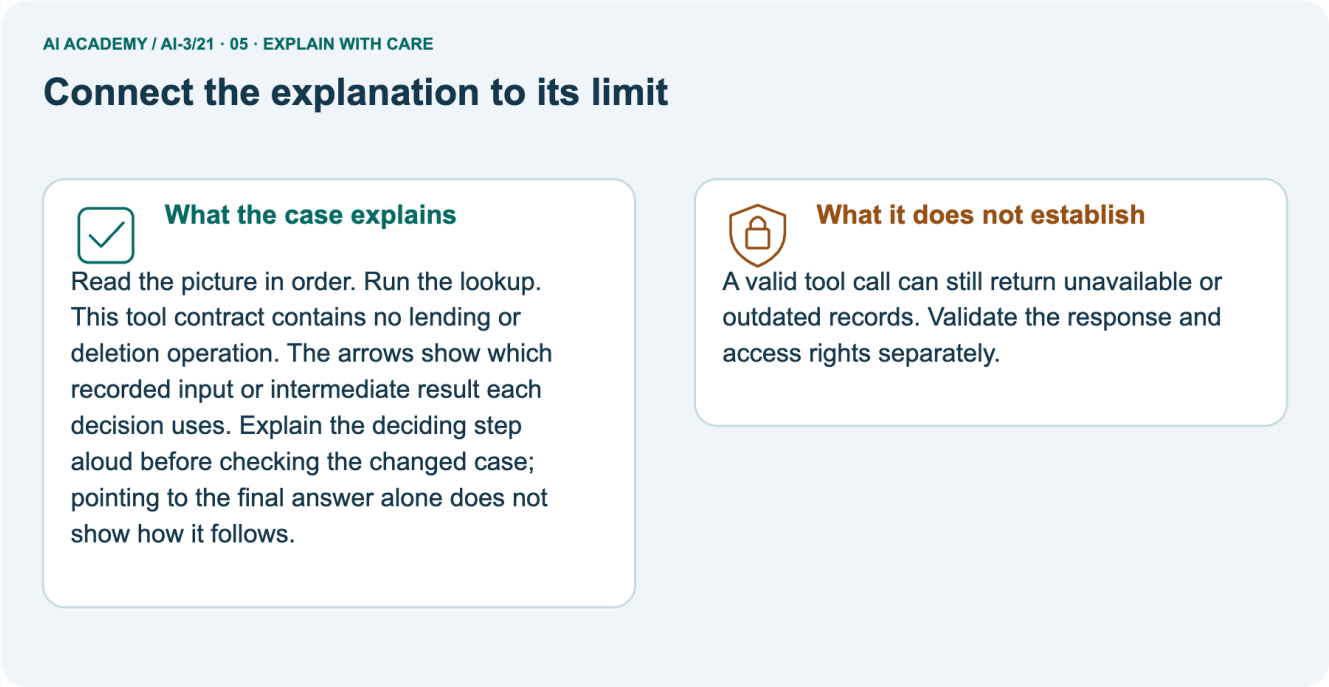
Reason: An extra operation is outside the exact allowed contract, even when book_id is valid.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: An extra operation is outside the exact allowed contract, even when book_id is valid.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Run the lookup. This tool contract contains no lending or deletion operation. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: A valid **tool call** can still return unavailable or outdated records.

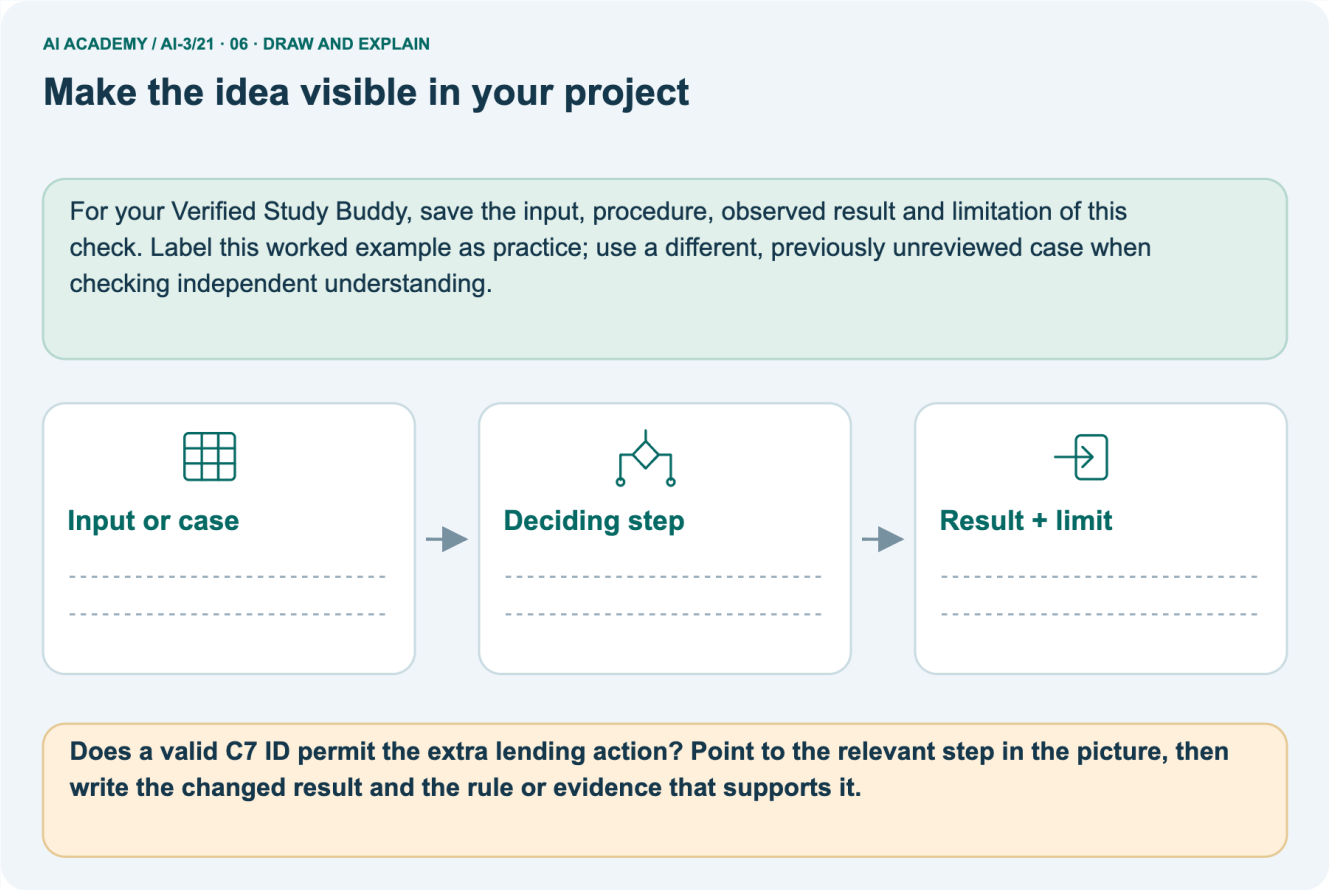
Validate the response and access rights separately.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A valid **tool call** can still return unavailable or outdated records. Validate the response and access rights separately.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does a valid C7 ID permit the extra lending action? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. ID validity and operation permission are separate checks; reject the extra field under this contract. Changed-case check: Reject the call before invoking the tool.

An extra operation is outside the exact allowed contract, even when book_id is valid.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Tools and APIs: the case

02. Follow the deciding steps

03. Tools and API contracts — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.